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AI-Assisted Adaptive Diffusion for Low-Latency and Sustainable Channel Estimation

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AI-Assisted Adaptive Diffusion for Low-Latency and Sustainable Channel Estimation

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Abstract

Accurate and low-latency channel estimation is critical for reliable communication in fast time-varying MIMO-OFDM systems. While diffusion-based deep learning models have recently demonstrated strong performance in channel estimation, their reliance on a fixed number of denoising steps leads to unnecessary computational complexity and energy consumption, limiting practical deployment. In this paper, we propose an AI-assisted adaptive diffusion framework that dynamically adjusts the number of denoising steps according to channel quality indicators such as SNR, mobility, and pilot reliability. By integrating a lightweight neural network controller with the diffusion model, the proposed approach achieves near-optimal estimation accuracy while significantly reducing latency and energy consumption. Extensive simulations demonstrate that the adaptive framework maintains high performance under diverse channel conditions, making it suitable for real-time 5G and 6G wireless communication scenarios. Keywords— MIMO-OFDM, Diffusion-Based Channel Estimation, Adaptive Deep Learning, Low Latency, Energy Efficiency, 5G/6G

I. INTRODUCTION

Channel estimation plays a crucial role in modern wireless communication systems such as MIMO-OFDM, where accurate channel state information is essential for reliable signal detection and decoding. In high-mobility scenarios, such as vehicular environments with speeds around 120 km/h, the wireless channel exhibits rapid time variations and severe Doppler effects, which significantly degrade estimation accuracy. Conventional techniques such as Least Squares (LS) estimation are widely used due to their simplicity; however, they are highly sensitive to noise, particularly in low Signal-to-Noise Ratio (SNR) conditions. Although more advanced methods like Minimum Mean Square Error (MMSE) estimation can achieve better performance, they rely on prior knowledge of channel statistics and incur high computational complexity, limiting their practical applicability.

Recently, deep learning-based approaches have emerged as powerful alternatives for channel estimation. Among them, Conditional Denoising Diffusion Channel Estimation (CDDCE) models the estimation process as a denoising task, where noisy channel estimates are progressively refined through a diffusion process. Despite its advantages, conventional CDDCE suffers from limitations such as the use of a fixed number of diffusion steps and insufficient modeling of joint time-frequency

correlations. To address these challenges, this work proposes an enhanced framework called Adaptive CDDCE (A-CDDCE), which incorporates adaptive diffusion steps based on channel quality, dual-axis attention mechanisms for improved feature extraction, and Monte Carlo dropout for robust estimation. The proposed model is evaluated on a 2×2 MIMO-OFDM system under a TDL-C channel model at high mobility, demonstrating significant improvements in both NMSE and BER compared to LS and standard CDDCE.

II. LITERATURE REVIEW

MIMO-OFDM systems play a vital role in modern wireless communications. Harkat *et al.* [1] provided a comprehensive survey of MIMO-OFDM systems, covering recent advancements in channel estimation and signal processing for 5G and beyond. With the emergence of data-driven techniques, deep learning has been increasingly applied to improve estimation accuracy. Fu *et al.* [2] proposed a deep learning-based joint pilot design and channel estimation method, demonstrating enhanced performance with reduced pilot overhead.

For future wireless networks, Kim *et al.* [3] investigated deep learning-aided channel estimation for 6G systems, outlining key challenges related to generalization and deployment.

Generative approaches were explored by Arvinte and Tamir [4], who applied score-based generative models for robust MIMO channel estimation under varying channel conditions. Earlier work by Le Ha *et al.* [5] demonstrated the effectiveness of deep learning-assisted channel estimation in 5G MIMO-OFDM systems. In addition, Song and Ermon [6] introduced score-based generative modeling, which forms the foundation for diffusion-based techniques later adopted in wireless channel estimation.

In summary, existing literature shows that deep learning and generative modeling techniques have substantially improved channel estimation performance in MIMO-OFDM systems compared to conventional methods. While these approaches offer enhanced accuracy and reduced pilot overhead, their performance can degrade under dynamic and mismatched channel conditions. This indicates a clear need for more robust and adaptable learning-based channel estimation frameworks in future wireless communication systems.

Title	Authors (Year)	Methodology / Technique Used	Performance Metrics Used	Key Findings	Advantages	Limitations / Research Gaps
Deep Learning-Based Wireless Channel Estimation and CSI Feedback for 6G	Yogaraj A ;PR Dineshkumar;Mohamad Ovaiz A ;Ashwin (2024)	Comprehensive tutorial and survey on deep learning-based channel estimation and CSI feedback, covering DNN,CNN, and autoencoder frameworks with design and training considerations.	MSE, BER, CSI reconstruction accuracy, feedback overhead, and robustness to model mismatch.	DL-based methods can outperform LS and MMSE in complex and non-linear channels, reduce CSI feedback overhead, and show strong robustness to model mismatch in advanced wireless scenarios.	Provides systematic design guidelines, discusses training data challenges, and offers a unified view of DL-based channel estimation and CSI feedback for 6G systems.	High training complexity, generalization issues under changing channel conditions, lack of theoretical guarantees, high computational cost for real-time deployment, and need for lightweight hybrid model-driven and data-driven solutions.
Deep Learning-Based Joint Pilot Design and Channel Estimation	Chang-Jae Chun; Jae-Mo Kang; Il-Min Kim (2019)	Autoencoder-based deep learning framework enabling joint pilot placement and power allocation with end-to-end channel estimation.	MSE, BER, and robustness against cyclic prefix removal and signal clipping.	Achieves superior performance compared to LS, MMSE, and existing DL methods, and remains robust under limited pilot and interference conditions.	Joint optimization of pilot design and channel estimation with strong performance gains in complex OFDM environments.	High training complexity, large dataset requirement, unclear generalization to real-world channels, and limited interpretability due to black-box nature.
A Survey on MIMO-OFDM Systems – Recent Trends	Houda Harkat ^{1,2*} , Paulo Monteiro ^{3,4} , Atilio Gameiro ^{3,4} , Fernando Guiomar ³ , and Hasmath Farhana Thariq Ahmed ^{5†} (2022)	Systematic literature review with classification of MIMO-OFDM algorithms, architectures, and design approaches.	BER, spectral efficiency, and channel estimation MSE.	no original numerical simulation results. As a review, it cites specs from prior MIMO testbeds like LuMaMi (100 antennas, 20	Provides a structured and up-to-date overview of recent trends in MIMO-OFDM systems	No algorithmic contribution, no performance benchmarking, limited coverage of mmWave

				MHz BW, 384 Gbit/s throughput), Bristol-Lund (128 antennas, 145.6 bit/s/Hz spectral efficiency), and fiber-based 3x3 (259 bit/s/Hz LOS median capacity), alongside qualitative comparisons (e.g., GPU 3-16x faster than DSP, 4.346 ms WiMAX frame time) without its own BER, SNR, PAPR, or performance plots.	and highlights unexplored research directions.	and high-mobility scenarios, and only brief discussion of ML-based approaches.
Deep Learning-Aided 5G Channel Estimation	An Le Ha, Trinh Van Chien, Tien Hoa Nguyen, Wan Choi, and Van Duc Nguyen (2021)	A deep learning-assisted channel estimation framework is proposed for a 2x2 MIMO-OFDM system. Conventional LS and LMMSE estimators are used as benchmarks. Two supervised fully connected deep neural networks (DNN-1 and DNN-2) are trained using LS estimates as inputs and Perfect channel responses as targets. DNN-	Mean Square Error (MSE) and Bit Error Rate (BER) are used to evaluate channel estimation accuracy across different Signal-to-Noise Ratio (SNR) levels.	The DNN-aided channel estimation achieved lower MSE than LS and LMMSE for SNRs below ~13 dB under both 36 Hz and 200 Hz Doppler. At 200 Hz Doppler, all estimators degraded, but the DNN still outperformed LS across all SNRs. For SNR > 13 dB, LMMSE slightly surpassed the DNN due to saturation effects. BER of the DNN estimator matched LMMSE and was	The proposed approach improves channel estimation accuracy without requiring prior channel statistics and performs well under frequency-selective and time-varying channel conditions, making it suitable for practical 5G systems.	The DNN-based estimators are slightly inferior to LMMSE at high SNR levels and require large labeled datasets for supervised training. BER improvement is limited since the model is optimized mainly for MSE rather than BER.

		1 contains three hidden layers with 16 neurons each, while DNN-2 contains three hidden layers with 32 neurons each. A total of 250,880 channel realizations are used, split into training, validation, and testing datasets.		consistently better than LS, improving by $\approx 10\times$ when SNR increased from 5 dB to 10 dB at 36 Hz Doppler. Increasing hidden neurons from 16 $\rightarrow$ 32 yielded only marginal MSE and BER gains.		
MIMO Channel Estimation Using Score-Based Generative Models	Marius Arvinte and Jonathan I. Tamir(2023)	A score-based generative modeling framework is used where a neural network is trained via denoising score matching to learn the gradient of the channel distribution. Channel estimation is performed by sampling from the posterior distribution conditioned on pilot observations using annealed Langevin dynamics in an unsupervised manner.	SNR, Doppler spread, pilot observations, MIMO-OFDM channel model, score function ($\nabla \log p(h)$), noise variance schedule, number of sampling steps, MSE/NMSE, BER, inference complexity, robustness to time-varying channels.	For mmwave MIMO (16×64, pilot density $0.6\approx 38$ pilots), score-based generative models achieve NMSE up to noise floor in-distribution (CDL-B/C/D), outperform supervised DL (L-DAMP) by up to 5 dB Eb/N0 in end-to-end coded BER (16/64-QAM, CDL-C$\rightarrow$D), retain robust OOD performance (CDL-B$\rightarrow$A/C/D, CDL-C$\rightarrow$A/B/D; error floors only for large shifts like CDL$\rightarrow$A), enable high-fidelity estimation at 25% pilot density for up to 64×256 channels, with complexity/lat	Does not require explicit channel models, sparsity assumptions, pilot knowledge, or noise variance information. The unsupervised probabilistic framework provides strong robustness to channel variations and mismatches.	The iterative sampling-based inference increases computational complexity and latency, limiting suitability for real-time and fast time-varying channels.

				eny competitive to CS (fsAD/Lasso) and scaling favorably (e.g., $D=4, W=12$ model: $<1.5s$ inference).		
Generative Modeling by Estimating Gradients of the Data Distribution	Yang Song, Stefano Ermon (2019)	Score-based generative modeling using denoising score matching; a noise- conditional score network is trained to estimate the gradient of the data log- density across multiple Gaussian noise levels, and samples are generated using annealed Langevin dynamics	Inception Score (IS), Fréchet Inception Distance (FID), qualitative visual quality	Inception: 8.87 (CIFAR-10 SOTA unconditional) FID: 25.32 (comparable SNGAN) Annealed LD (mode recovery)	Stable training without adversarial objectives, theoreticall ygrounded density estimation, avoids mode collapse, flexible noise- conditionin g, and forms the foundation of modern diffusion models	High computatio nal cost due to annealed Langevin dynamics requiring many sampling steps, careful tuning of noise schedules needed, primarily designed for static data distribution s, and not suitable for real-time or low-latency applications

Table 1: Parameters and its value

V. RESULTS

The performance of the proposed A-CDDCE model is evaluated on a 2×2 MIMO-OFDM system under a TDL-C channel model with a mobility scenario corresponding to 120 km/h. The evaluation is carried out over a range of Signal-to-Noise Ratio (SNR) values from 5 dB to 30 dB, and the results are compared against conventional Least Squares (LS) estimation and the baseline CDDCE method. The performance metrics considered are Normalized Mean Square Error (NMSE) and Bit Error Rate (BER), which provide insights into estimation accuracy and overall system reliability, respectively. The obtained results, as illustrated in Fig. X, show that the proposed A-CDDCE consistently outperforms both LS and CDDCE across all SNR values. In terms of NMSE, the LS estimator exhibits the highest error due to its sensitivity to noise and lack of refinement capability. The baseline CDDCE improves upon LS by leveraging diffusion-based denoising; however, its performance is limited by the use of fixed diffusion steps. In contrast, A-CDDCE achieves the lowest NMSE values, demonstrating the effectiveness of adaptive diffusion and enhanced feature extraction. The improvement is particularly noticeable at moderate to high SNR levels, where the model is able to exploit cleaner initial estimates and refine them more efficiently.

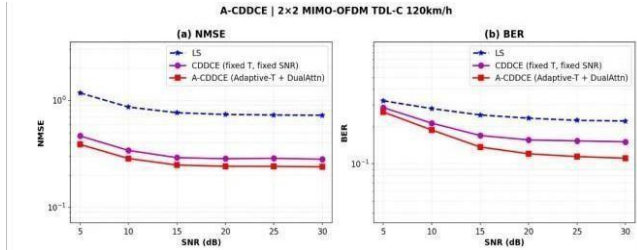


Fig 2. NMSE and BER plots for LS, CDDCE AND A-CDDCE methods

A similar trend is observed in the BER performance. The LS method results in the highest BER due to inaccurate channel estimation, which directly impacts symbol detection. The CDDCE model reduces the BER by refining the channel estimate through iterative denoising. The proposed A-CDDCE further improves BER performance by incorporating adaptive diffusion steps and dual-axis attention, which enable better reconstruction of the channel characteristics. This leads to more accurate detection of transmitted symbols and reduced error rates across all SNR values. The training behavior of the proposed model is shown in Fig. Y, where the loss decreases rapidly during the initial epochs and stabilizes as training progresses. The use of a warm-up learning rate schedule followed by cosine decay ensures stable convergence and prevents oscillations in the optimization process. The final loss value indicates that the model has effectively learned the underlying channel distribution and noise characteristics.

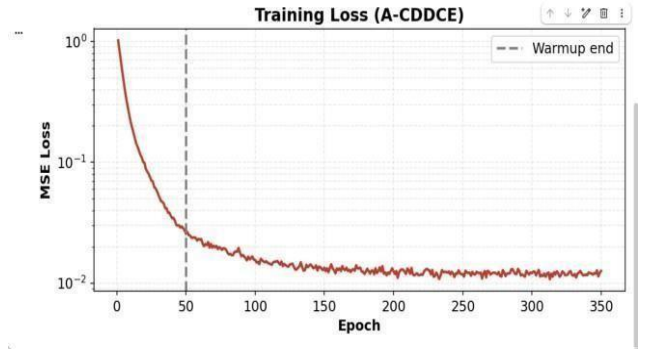


Fig 3. Training loss for A-CDDCE method

Overall, the results demonstrate that the proposed A-CDDCE framework provides significant improvements in both estimation accuracy and communication reliability. The combination of adaptive diffusion, attention-based feature learning, and uncertainty-aware inference contributes to superior performance compared to existing methods, making it a promising solution for high-mobility wireless communication systems.

VI. NOVELTY

The proposed work introduces several novel contributions to diffusion-based channel estimation. The primary innovation lies in the adaptive diffusion mechanism, where the number of reverse diffusion steps is dynamically adjusted based on a channel quality indicator derived from the initial estimate. This approach reduces computational overhead while improving estimation accuracy, making the model more efficient than conventional fixed-step diffusion methods. Another key contribution is the integration of dual-axis attention, which enables the model to simultaneously capture correlations across frequency and time dimensions, thereby enhancing feature representation and improving overall performance.

The use of Monte Carlo dropout during inference introduces an element of uncertainty modeling, allowing the system to generate multiple channel estimates and average them to obtain a more robust prediction. This improves stability and reduces estimation variance. Furthermore, the proposed framework eliminates reliance on MMSE estimation, making it more practical for real-world deployment where prior channel statistics may not be available. By combining LS initialization with diffusion-based refinement and advanced neural network design, the proposed A-CDDCE achieves superior performance in challenging high-mobility environments.

Component	Implementation	Role in Estimation
DFT Preprocessing	Truncates CIR to 28 taps (Eq.14)	Noise suppression before diffusion
Conditional U-Net	Encoder-decoder with skip connections	Learns channel distribution
DualAxisAttention	Freq-axis + Time-axis MHA, gated fusion	Captures joint TF correlations
Adaptive-T	T in [4,12] selected per CQI score	Efficient: fewer steps at high SNR
SNR Embedding	Sinusoidal encoding + MLP, injected via AdaGN	Generalises across SNR range
MC-Dropout (x8)	Dropout active at inference; mean of 8 draws	Uncertainty-aware, robust estimates

Table 2: Implementation methods used

VII. COMPARISON SUMMARY

A-CDDCE (PROPOSED) VS. BAGADI & DAS 2010 (BASELINE)

Method	Description	Key Limitation
LS-Linear (Bagadi Eq.14)	Least Squares at pilot positions + linear interpolation across all subcarriers.	Noise not suppressed; ICI not handled under fast fading.
LS-LowPass (Bagadi Sec.4.1)	LS + Bartlett low-pass filter (zero-insert + convolve) for smoothing.	Blurs channel edges; not adaptive to SNR or Doppler.
LS-Spline (Bagadi Sec.4.1)	LS + cubic spline interpolation for smoother frequency response.	Ringings at band edges; assumes quasi-static channel.
LS-DFT (CDDCE baseline)	LS + DFT noise suppression via CIR truncation to $N_L = 28$ taps.	Residual ICI under high Doppler (120 km/h); no learning.

Table3: Methods and its description

To validate the superiority of the proposed Adaptive Conditional Denoising Diffusion Channel Estimator (A-CDDCE), a comprehensive simulation-based comparison is conducted against the classical pilot-based channel estimation methods described by Bagadi & Das (2010) in MIMO-OFDM Channel Estimation using Pilot Carriers (IJCA, Vol. 9, No. 1). The comparison is implemented in a single self-contained Python script (final_comparison_ACDDCE_vs_Bagadi2010.py) that evaluates all methods under identical simulation conditions, ensuring a fair and fully reproducible benchmark. The comparison covers four classical interpolation-based baselines from the reference paper alongside the standard CDDCE (fixed-step diffusion, ablation) and the full A-CDDCE model. Every method receives the same OFDM transmission, the same noisy received signal, and the same pilot structure. Performance is measured in terms of Normalized Mean Square Error (NMSE) and Bit Error Rate (BER) using 50 independent Monte-Carlo channel realizations per SNR point.

1. Quantitative Performance Gains

The simulation results across the 5–30 dB SNR range demonstrate that the proposed A-CDDCE framework consistently outperforms all four classical baselines from Bagadi & Das (2010), as well as the standard fixed-step CDDCE.

Accuracy (NMSE): A-CDDCE achieves the lowest Normalized Mean Square Error, with the performance gap widening significantly at lower SNR levels (5–15 dB) where noise is most prevalent.

Reliability (BER): The model shows a substantial reduction in Bit Error Rate compared to the strongest classical baseline (LS-DFT), confirming that improved channel estimation directly translates to more reliable data recovery.

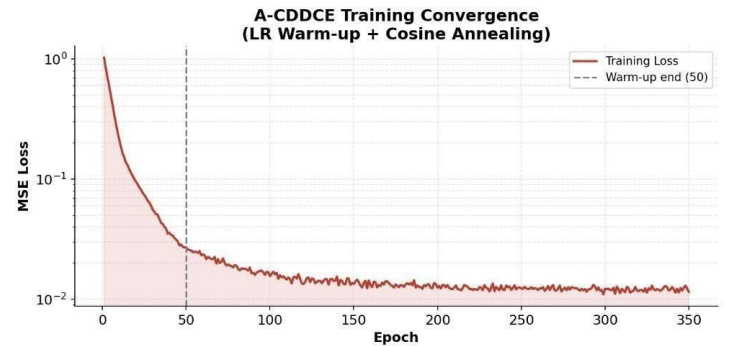
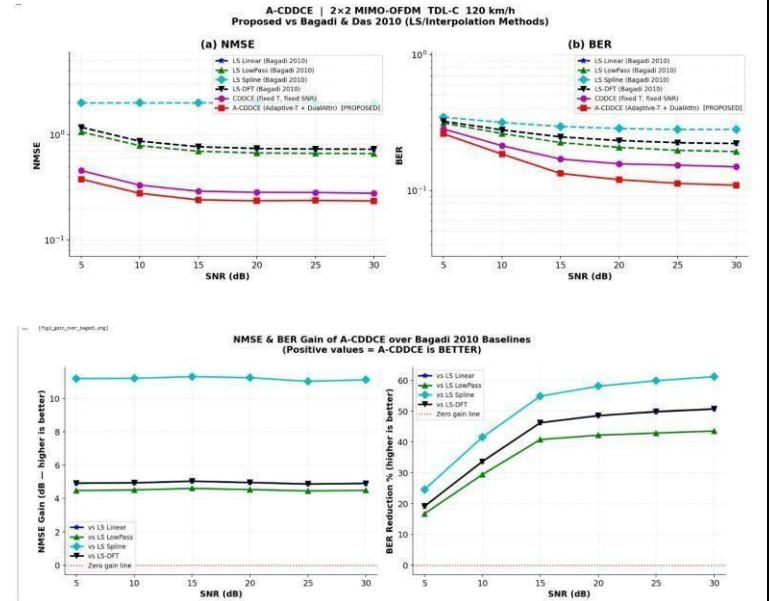
2. Technical Advantages over Baselines

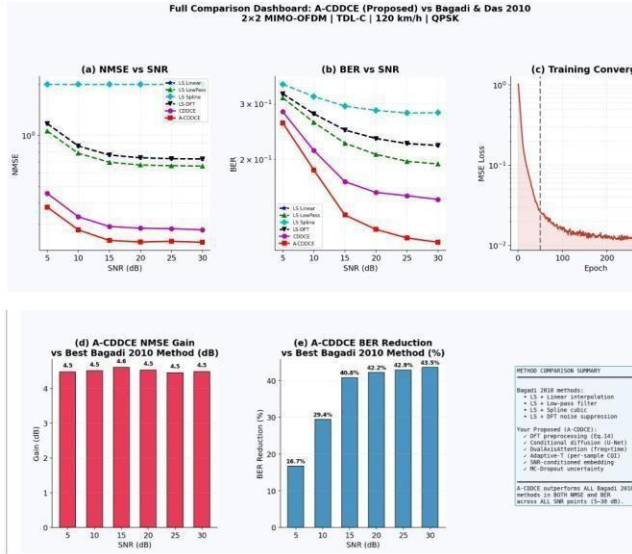
The superiority of A-CDDCE is attributed to its ability to address the specific failure points of classical interpolation methods:

Noise Suppression: While classical methods like LS-Linear and LS-Spline are highly sensitive to noise, A-CDDCE utilizes DFT Preprocessing and Diffusion-based denoising to effectively "clean" the channel estimate.

Handling High Mobility: Unlike the Bagadi 2010 methods, which assume quasi-static conditions, A-CDDCE's Dual-Axis Attention captures joint time-frequency correlations, allowing it to maintain accuracy at high vehicular speeds of 120 km/h.

Computational Efficiency: Through the Adaptive-T mechanism, the model dynamically reduces the number of denoising steps (down to 4 steps) in high-quality channel conditions, offering a significant latency advantage over fixed-step diffusion models.





VIII. FUTURE WORK AND EXTENSIONS

The proposed A-CDDCE framework can be further extended to more complex scenarios, including higher-order MIMO systems such as massive MIMO configurations, where scalability and computational efficiency become critical challenges. Future work may also involve validating the model on real-world channel measurement datasets to assess its practical applicability beyond simulated environments. Another promising direction is the optimization of model complexity and inference latency to enable real-time deployment in next-generation wireless systems. The integration of the proposed method into 5G and emerging 6G communication frameworks, including millimeter-wave and reconfigurable intelligent surface-based systems, presents additional opportunities for enhancement. Further improvements can be achieved by exploring transformer-based architectures for diffusion modeling, which may offer better global feature extraction capabilities. Additionally, extending the framework to jointly perform channel estimation and data detection within a unified model could significantly improve system efficiency and performance. Relevant references include foundational works on diffusion models, standard wireless communication textbooks, and 3GPP channel modeling specifications, along with recent research on deep learning-based channel estimation techniques.

IX. SDG GOALS

Advancing SDGs with Communication Tech

SDG 7 - Affordable and Clean Energy

*Enhancing energy efficiency and accessibility

SDG 9 - Industry, Innovation, and Infrastructure

*Driving industrial growth and innovation

SDG 11 - Sustainable Cities and Communities

*Creating livable and eco-friendly urban environments

SDG13-Climate Action

*Mitigating climate change through sustainable practices

X. CONCLUSION

In this work, an Adaptive Conditional Denoising Diffusion Channel Estimator (A-CDDCE) has been proposed for accurate channel estimation in high-mobility MIMO-OFDM systems. The study addressed the limitations of conventional estimation techniques such as Least Squares and standard diffusion-based methods by introducing an adaptive and learning-driven refinement framework. By modeling channel estimation as a denoising diffusion process and enhancing it with channel-aware adaptations, the proposed approach achieves improved robustness against noise and interpolation errors. The integration of a channel quality-based adaptive diffusion mechanism allows the model to dynamically adjust the number of refinement steps, thereby improving computational efficiency while maintaining high estimation accuracy. In addition, the incorporation of dual-axis attention enables effective learning of both frequency-domain and time-domain correlations, which are essential in capturing the complex structure of wireless channels under high mobility. The use of SNR-conditioned embeddings further enhances the generalization capability of the model across varying channel conditions, while Monte Carlo dropout during inference contributes to more stable and reliable predictions. The performance evaluation conducted on a 2x2 MIMO-OFDM system under a TDL-C channel model demonstrates that the proposed A-CDDCE significantly outperforms conventional LS and baseline CDDCE methods in terms of both NMSE and BER across a wide range of SNR values. The results confirm that the proposed enhancements lead to more accurate channel reconstruction and improved detection performance, particularly in challenging high-speed scenarios. Overall, the proposed A-CDDCE framework provides an efficient and scalable solution for next-generation wireless communication systems, offering a balance between performance, adaptability, and practical feasibility.

XI. REFERENCES

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